



Marine Biology

海洋生物学



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Course Outline :

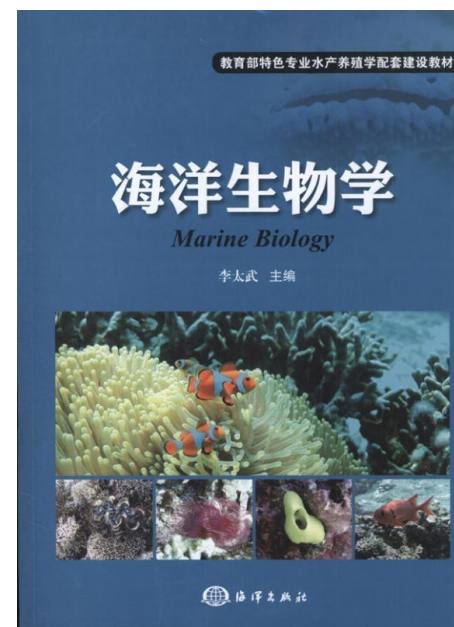
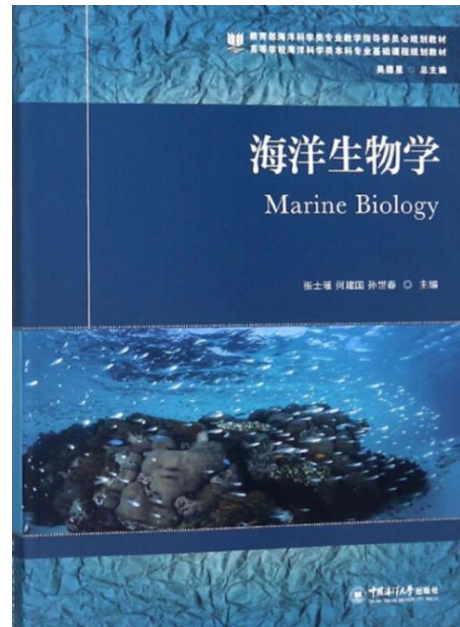
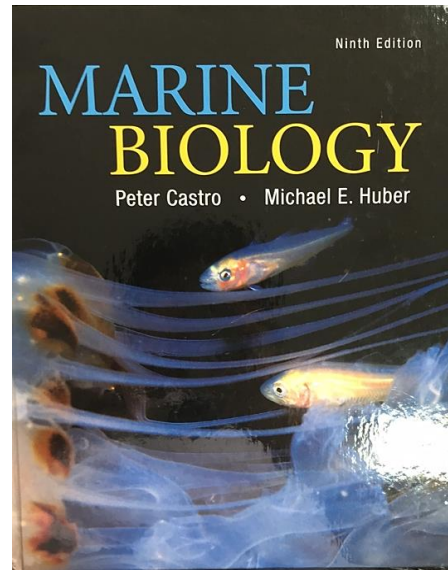
1. Course Plan:

- 48 courses
(32 theory + 16 practice)
- 3 grade points
- 2 terms (Autumn & Winter)

3. Course Contents:

- 10 Chapters
- Autumn: Chapter 1-5
- Winter: Chapter 6-10

2. Textbook:



浙江大学2024-2025学年秋冬学期校历

	暑假						秋学期								冬学期								寒假			
	七月	八月					九月				十月				十一月				十二月				一月			
							一	二	三	四	五	六	七	八	一	二	三	四	五	六	七	八	考试周			
星期一	29	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13	20
星期二	30	6	13	20	27	3	10	17	24	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21
星期三	31	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22
星期四	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23
星期五	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22	29	6	13	20	27	3	10	17	24
星期六	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25
星期日	4	11	18	25	1	8	15	22	29	6	13	20	27	3	10	17	24	1	8	15	22	29	5	12	19	26

Theories Practices

- Practice starting from the **Fifth week (10th Oct.)**
- Including at least 10 lab-based practices
- Detailed information about practice will be given before the practice, e.g. Tuesday's lecture
- Lecture note, homework and all related tasks will be published and submit via 'Learning in ZJU'

Course Criteria:

Part 1: Classwork & Homework **10%**

- Classwork/Quize—Key points in the lecture
- Homework—Creative thinking after the lecture

Part 2: Practice **40%**

- In the practice—skills, disciplines and thinking abilities
- After the practice—reports, samples and answers
- Field observation—discussion or presentation (optional)

Part 3: Examination (Mid-term + Final term) **40%**

- **Close-book** Exam both in **English and Chinese**
- Multi-format (Choice Q, Fill-in Gap, T&F, Questions, etc.)

Part 4: General Performance **10%**

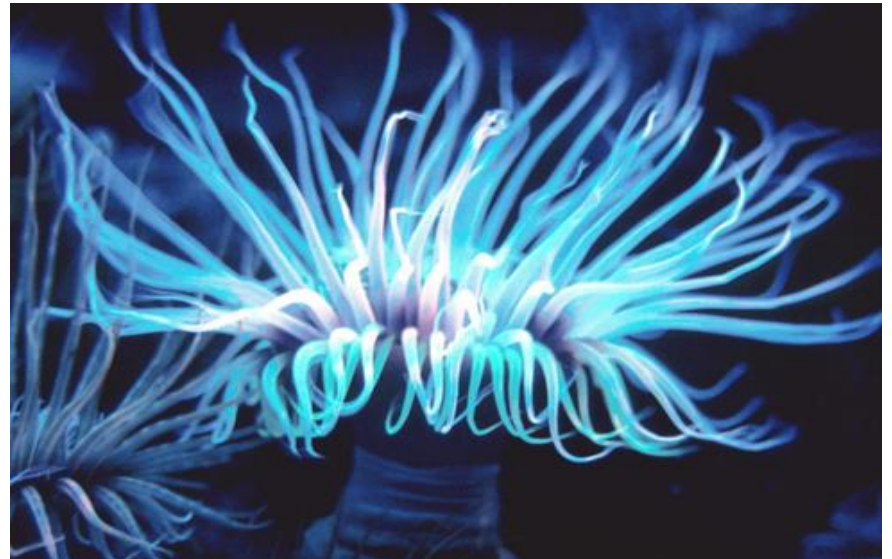
Attendance, question answering, participation, etc.



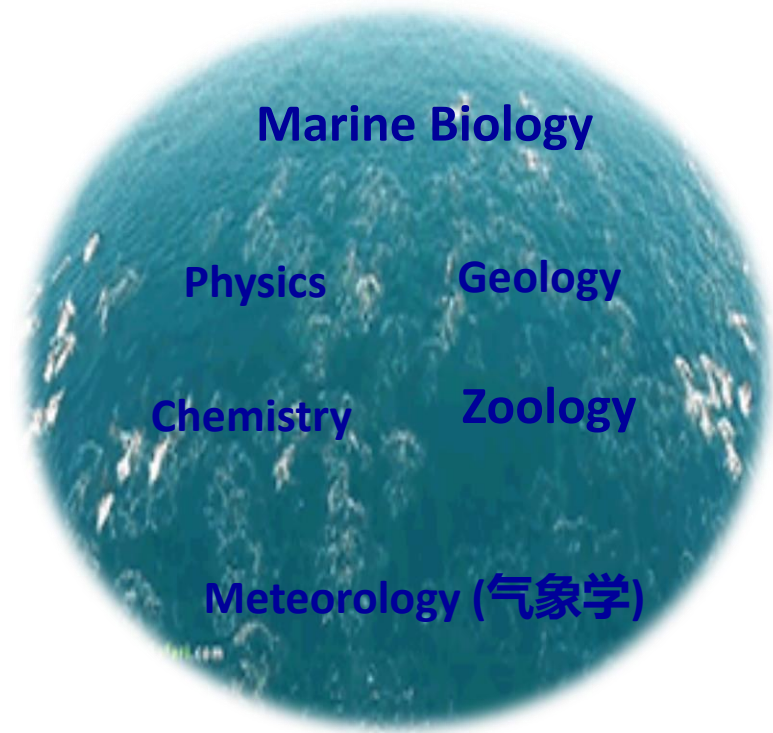
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The Science of Marine Biology

What is Marine Biology?



- **Marine biology** is the scientific study of **the abiotic and biotic characteristics of the ocean and their interactions**, including habitats of marine organisms, marine biodiversity, marine ecology and human impacts
- **Emphasis** is placed on the **variety of marine organisms and their structural, physiological, and behavioral adaptations within specific marine environments**
- **Special attention** is directed to **marine systems**, e.g., estuaries, coral reefs, and **factors** limiting the distribution of organisms within those systems
- Marine biology is **not a separate science**. It is an applied field of biology and incorporates many other sciences



Marine Biologist/Biological Oceanographer/Oceanographer

- **Marine Biologists** study the organisms that living close to the shore—**living things**
- **Biological Oceanographer** study the organisms that living in the open ocean—**living things**
- **Oceanographers** study the physical aspects of the ocean including tides, currents, waves and the chemical make-up of seawater—**non-living environment**

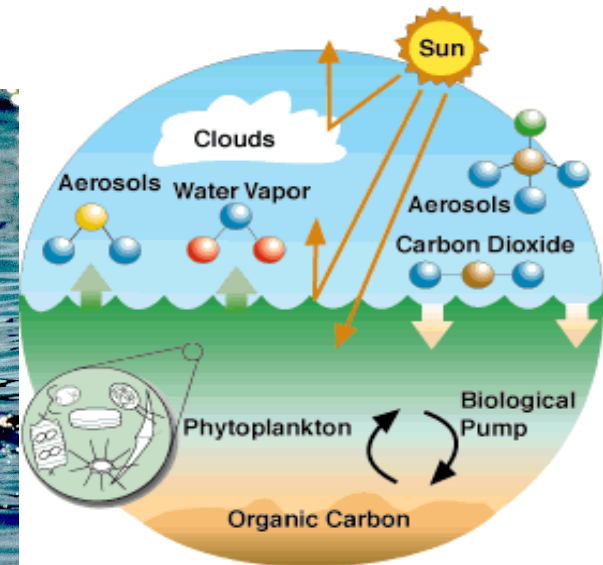
Why Study Marine Biology?

1. Life has arisen in the sea, studying marine life provides clues into the validity (**正确性**) of such claim
2. Many products come from the sea, such as medicines derived from marine species, food resources and other items used for human



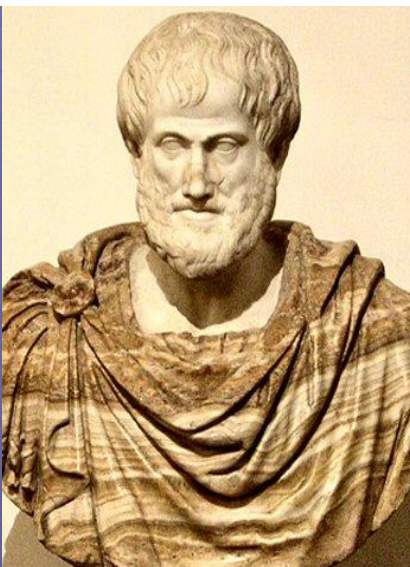
3. The marine environment and its habitats support recreation and tourism worldwide
4. Marine organisms produce oxygen used by aerobic organisms, including humans
5. Oceans help to regulate climate
6. Marine organisms can cause problems directly to human life or their property

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History of Marine Biology

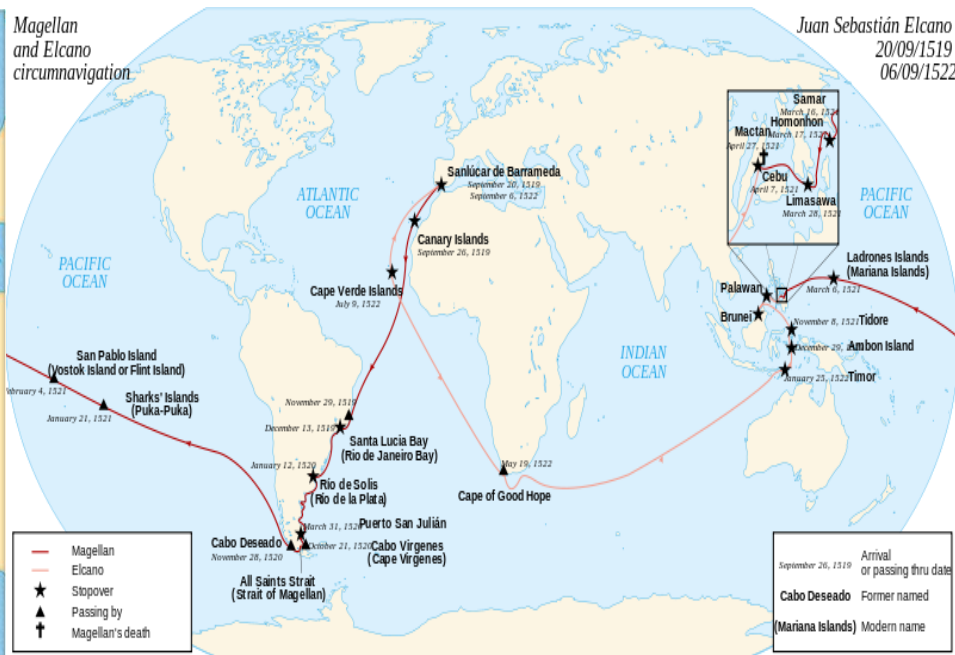
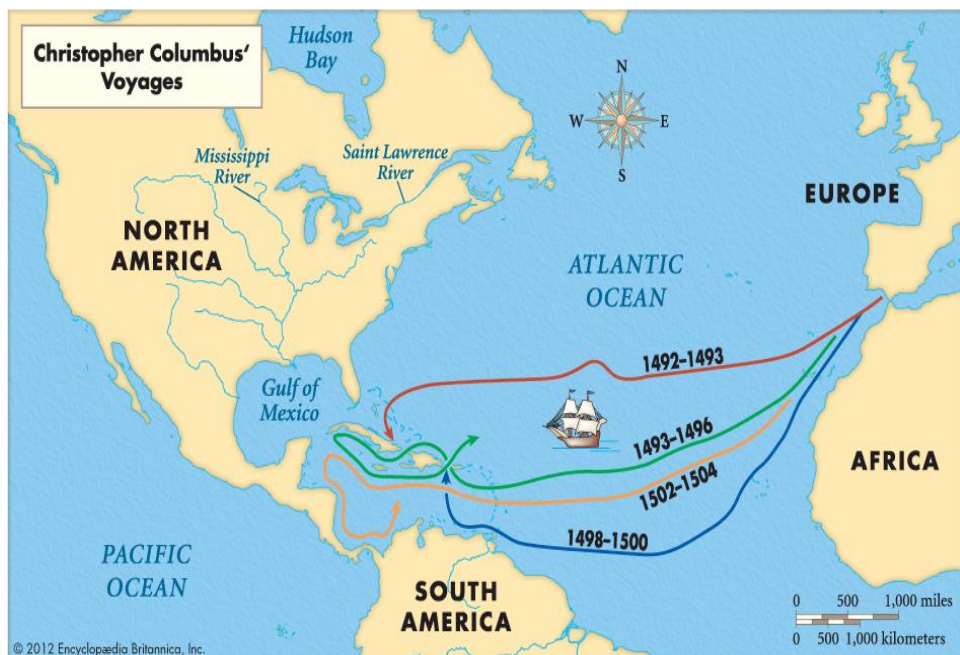
- Human use of the sea (>150,000 years old)
- Early ocean explorations by Pacific Islanders, **Phoenicians** (offshore) and **Ancient Greeks** (near shore). **Aristotle**, the Greek philosopher(哲学家), described many marine forms and their features



- Much exploration stopped during the **Dark Ages** (500-1000 AD), except for the explorations of the **Vikings** in the Ninth and Tenth Centuries. Vinland (文 兰) discovered (in 995 AD, still in doubt and debate)



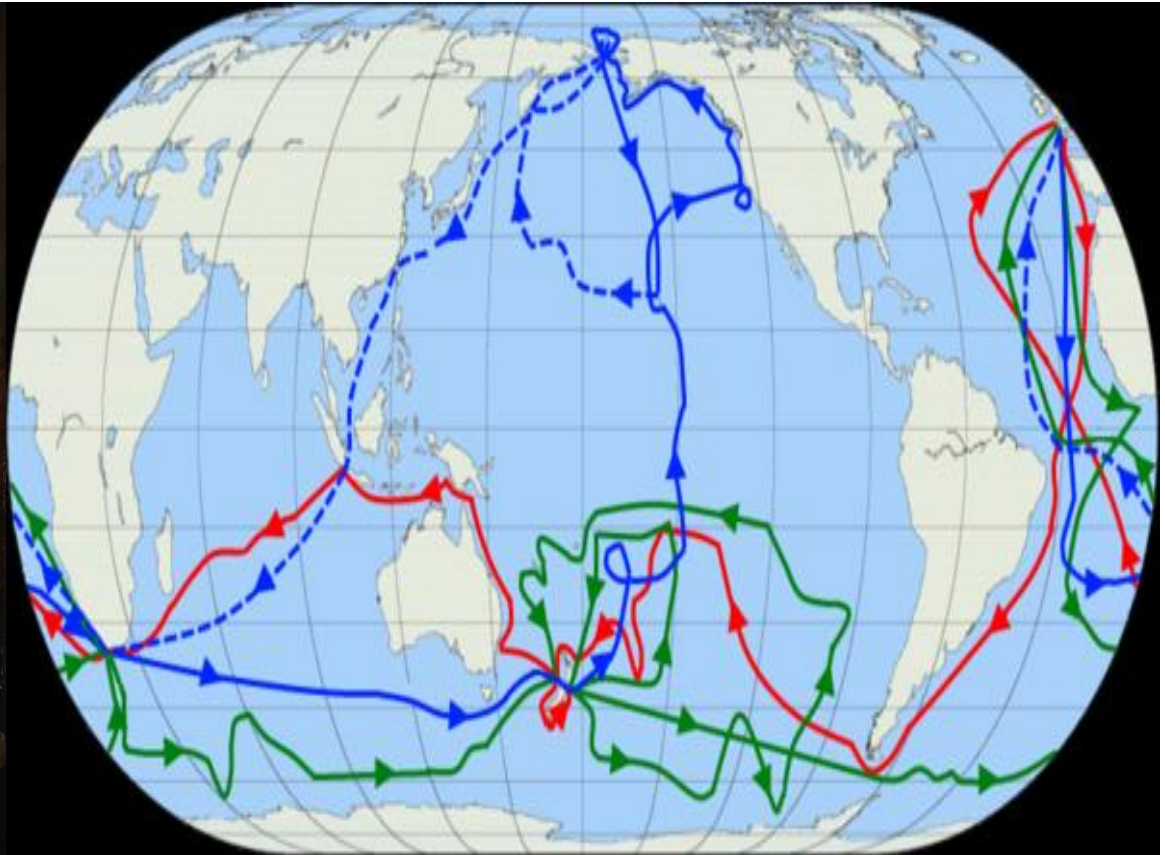
- Explorations began again during the Renaissance (文艺复兴时期):
 - **Columbus** “rediscovered” the “New World” or North America in 1492
 - **Magellan** sailed around the world in 1519



Juan Sebastián Elcano
20/09/1519
06/09/1522

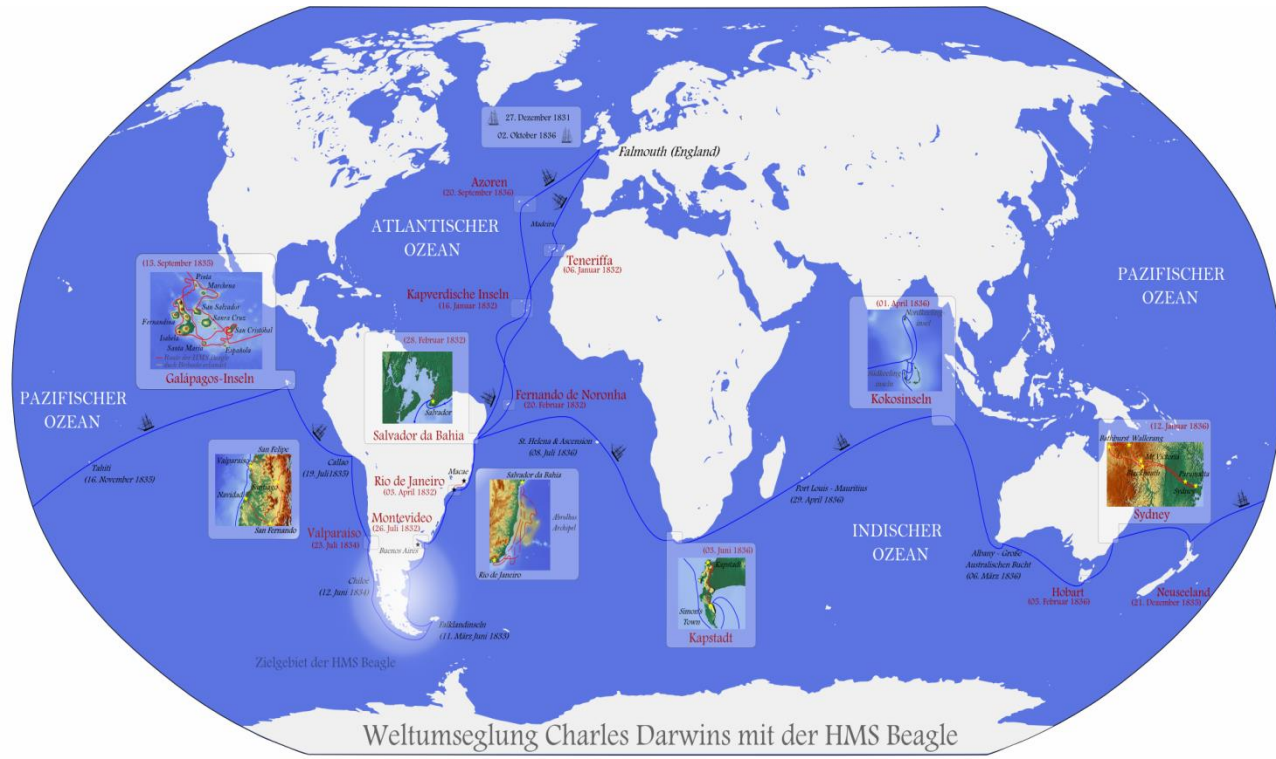
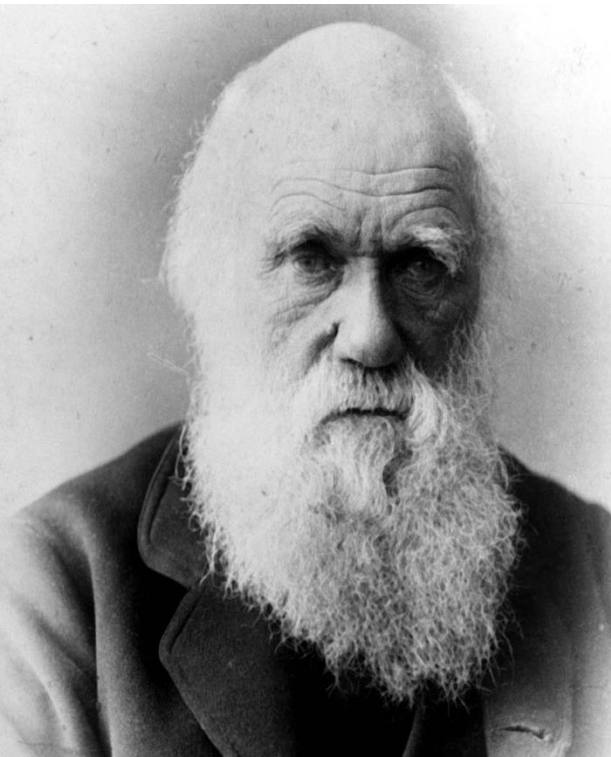
September 26, 1519 Arrival or passing thru date
Cabo Deseadó Former name
(Mariana Islands) Modern name

- **James Cook**, an English captain, and his crew sailed **all the oceans** and were the first European to view the Arctic ice fields, land on Hawaii, New Zealand, Tahiti as well as many other Pacific Islands



• Charles Darwin

- Sailed on the HMS Beagle from 1831-1836 as a naturalist (博物学家)
- The primary goal was to map coastlines
- He observed, collected and wrote detailed descriptions of the organisms (The voyage of the Beagle)

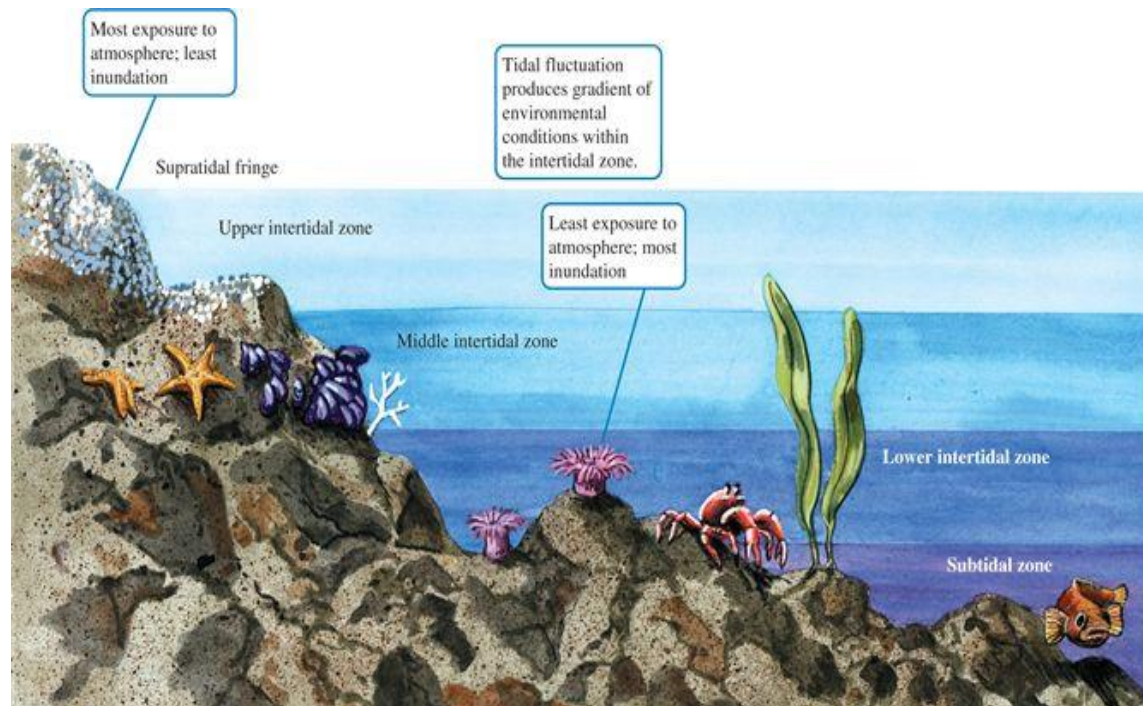


- **Edward Forbes**

- Extensively studied the **seafloor around the British Isles**, Aegean Sea and other locales in 1840's and 1850's
- He discovered that **species on the seafloor vary greatly depending on depth (Zonation)**

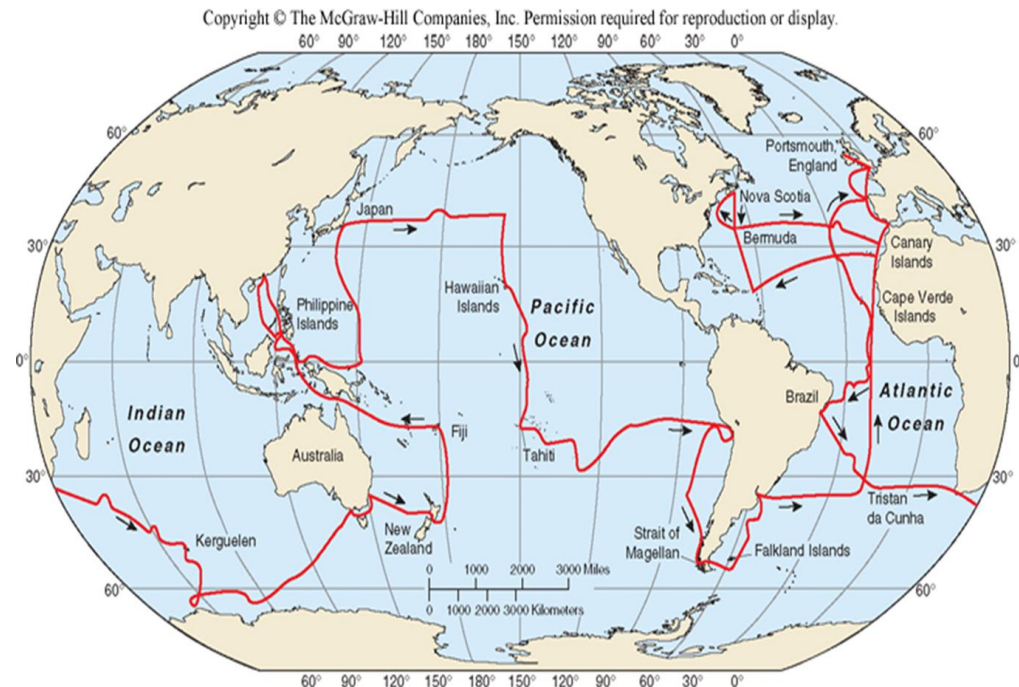
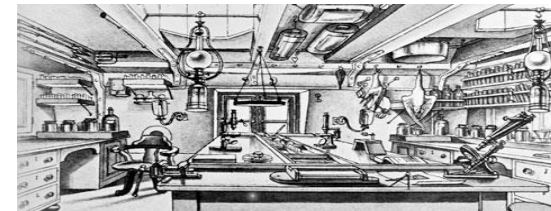


- **Zonation:** the presence of organisms in a particular range (or zone)



- **Charles Wyville Thompson**

- A scientific leader of the HMS *Challenger* Expedition that sailed from 1872-1876
- The **first major exploration devoted to studying marine organisms**
- The crew discovered thousands of species never previously described and published **50 volumes** of information from the data collected on the trip over the next **19 years**



- In China

- Sailing has long history for the expanding and trading, but seldom focused on discoveries in Marine Biology
- **Xu Fu's** Voyages to the east in Qin Dynasty
- **Maritime Silk Route**, formed since Xihan and still in developing currently
- **Jianzhen's** Voyages to the east (Japan, 6 times) in Tang Dynasty
- **Zheng He's** Voyages to the west (Atlantic ocean, Ming Dynasty, 1405-1433, 7 times), the first voyage to involve marine biology survey

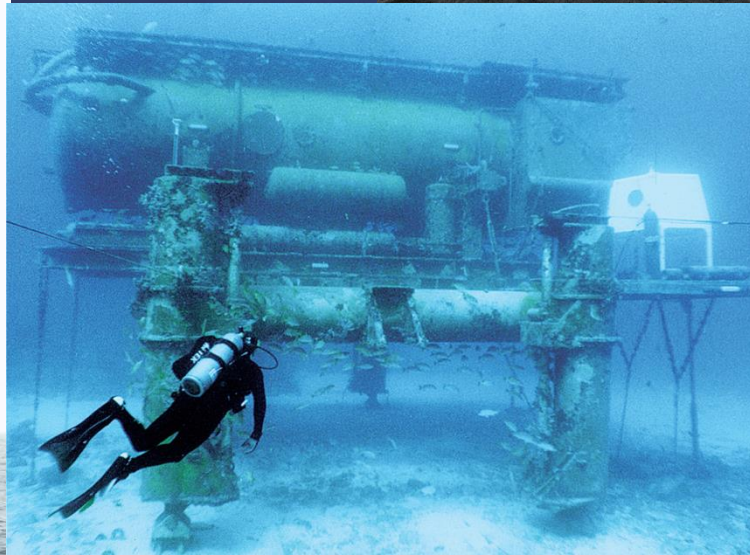
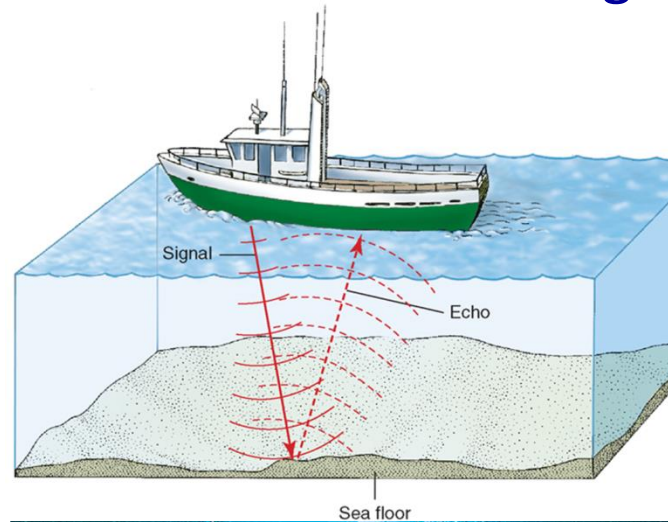
Modern Marine Biology

- **Today**, several marine biology research stations exist in locations around the world:
 - Stazione Zoologica, Naples, Italy
 - Woods Hole Marine Biological Lab, MA, USA
 - Scripps Institute - La Jolla, CA, USA
 - Friday Harbor Labs, WA, USA
 - Marine Biology Association, PL, UK
 - Commonwealth Scientific and Industrial Research Organization, WA, AUS

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Important Tools in Marine Biology

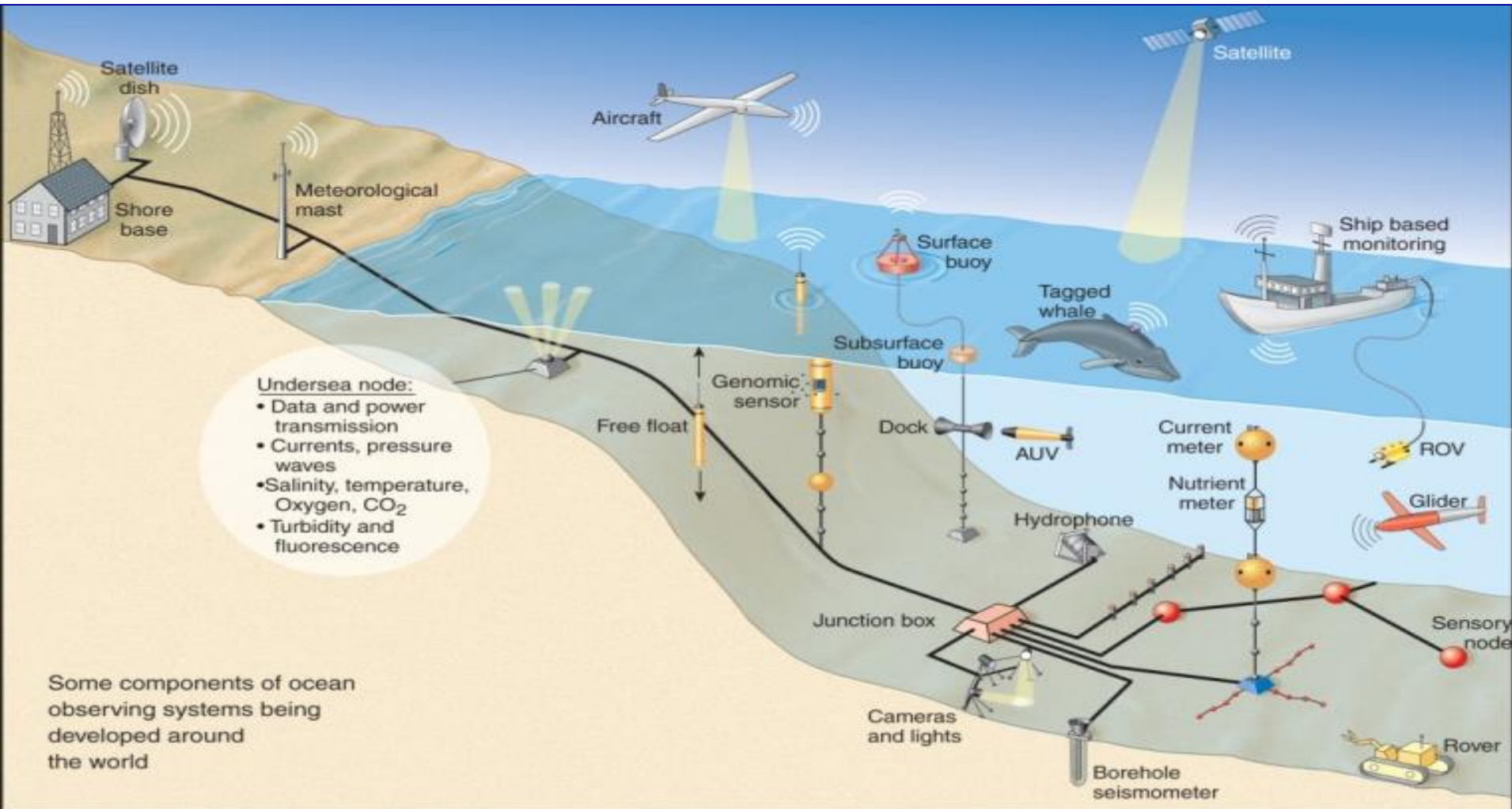
- Sonar
- Scuba
- Remote sensing
- Remotely operated vehicles (ROV's)
- Underwater Research Station - *Aquarius*
- Research vessels
- AI technology



Our tools



The North East Pacific Time-integrated Undersea Networked Experiments (NEPTUNE, 海王星) Observatory, which extends from British Columbia to Oregon



Pluto: 冥王星

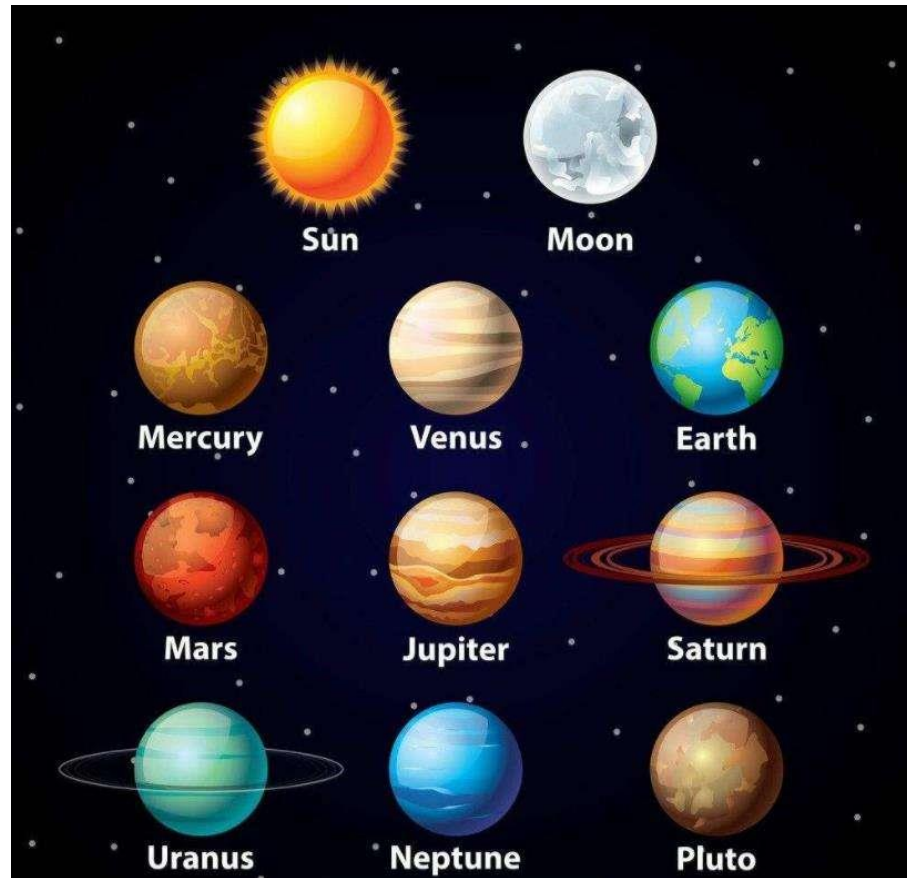
Uranus: 天王星

Jupiter: 木星

Mercury: 水星

Saturn: 土星

Earth: 地球



Venus: 金星 Victoria Experimental Network Under the Sea

Mars: 火星 Monterey Accelerated Research System